

2008 F=ma Exam: Problem 10

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By Newton's 2nd law, we have

$$F - f = ma$$

In terms of F ,

$$F = ma + f$$

so if we plot (a, F) data points, we will obtain a line with slope equal to the mass m . Choosing two of the data points,

$$m = \frac{\Delta F}{\Delta a} = \frac{5.05 \text{ N} - 3.05 \text{ N}}{0.495 \text{ m/s}^2 - 0.095 \text{ m/s}^2} = 5 \text{ kg}$$

so the answer is B.